

Structural Life Monitoring for Composite Gliders

Methodology Report

Based on: "Extension of Glider Life using Structural Life Monitoring" — J.L. Derouineau, FFVP / EGU [7]

1. Purpose and Scope

This document describes the methodology used to assess and justify extended operational life for composite gliders using Structural Life Monitoring (SLM). The method calculates an Equivalent Fatigue Index (EFI) (§4.4) from onboard acceleration measurements and compares the measured operational severity against the accepted certification reference severity represented by the Kossira & Reinke Utility reference spectrum.

The purpose is not to determine the absolute physical damage state of the composite structure. The purpose is to apply the same conservative fatigue-severity calculation to both the certification reference spectrum and the measured operational spectrum. As long as cumulative EFI remains below the certification reference limit, and the required inspection and continued-airworthiness provisions are satisfied, continued operation beyond the certified Service Life Limit may be justified without modifying Type Certificate documents or certified structural limits.

The methodology is applicable only to composite gliders operated within the EASA CS-22 Utility-category flight envelope (some glider models considered for SLM were initially certified under the National Authority of the TC holder, but all Type Certificate Data Sheets are now listed under the EASA product list), for which a Service Life Limit has been established. The method uses the Kossira & Reinke Utility reference load spectrum, recognized by EASA [8] (ABB.164). Aerobatic-category operation and any operation outside Utility-category limitations are excluded from the initial scope of SLM implementation.

2. Regulatory Basis

The method is consistent with the following regulatory framework:

- **CS 22.627** (Fatigue Strength): requires structures to be designed to avoid stress concentrations in normal service; does not explicitly mandate a specific life limit.
- **EASA CM-S-006**: endorses the use of Kossira & Reinke load spectra for fatigue analysis and testing of sailplanes.
- **CS-STAN 104b / Part ML**: governs installation of onboard recorders and maintenance requirements.

EASA has already accepted multiple means of compliance for CS 22.627, including full-scale fatigue testing and static overload testing. SLM is proposed as an additional means of compliance for ageing glider life management. Similar measured-usage principles have been used in aviation usage monitoring and Individual Aircraft Tracking, particularly in military aviation. SLM does not attempt to reproduce those military approaches directly or claim that they are directly transferable to gliders; the relevant precedent is the general principle that measured usage can be compared with a conservative reference usage assumption, provided the method is controlled, conservative, and supported by inspections.

3. Key Principles

Service Life Limit and Certification EFI Limit

The certified Service Life Limit (typically 12,000 flight hours) corresponds to an accepted fatigue exposure established during type certification. Certification is demonstrated against the safety-factored life: with a Safety Factor of 3, structural integrity is shown for 36,000 hours of equivalent fatigue exposure. In classical Palmgren-Miner notation, the cumulative value is often denoted D , for damage, and $D = 1$ is associated with the theoretical failure threshold. In this SLM methodology, the same mathematical quantity is denoted Equivalent Fatigue Index, or EFI , because it is used as a conservative usage-severity index rather than as a direct measurement of physical composite damage.

The reference condition for the SLM calculation is therefore:

$$EFI_{ref,36000} = 1$$

for the safety-factored 36,000-hour reference exposure, and:

$$EFI_{ref,12000} = \frac{1}{3}$$

for the certified 12,000-hour service life. The certification EFI limit used by the SLM method is:

$$EFI_{limit} = \frac{1}{3}$$

This does not mean that the actual composite structure is assumed to fail at 36,000 hours, nor that its physical damage state is known at 12,000 hours. For SLM, the accepted certification reference exposure is deliberately treated as the full allowable fatigue-exposure budget. This is conservative because the method does not take credit for any unobserved residual margin beyond the certified exposure.

SLM as a reference-spectrum comparison method

SLM is not certification by analysis and does not replace the structural substantiation performed during Type Certification. It does not attempt to determine the absolute fatigue life of a composite structure, identify the critical structural location, or predict the critical composite failure mode.

The method has a more limited objective: it uses the already accepted certified life exposure as a conservative reference and compares it with the measured operational exposure of an individual glider. The Palmgren-Miner calculation is therefore used as an Equivalent Fatigue Index, calculated consistently for the reference spectrum and for the measured spectrum.

Proportionality between load factor and structural stress

Within the Utility-category certified flight envelope — defined by speed, configuration (flaps, airbrakes), and authorized mass loading — load-factor variations are used as the practical measured proxy for the stress variations used in the EFI comparison. This proxy is justified by the proportionality, during aerodynamic flight, between vertical load factor and the bending-related structural stress variations that drive the reference Kossira & Reinke fatigue spectra. The manufacturer's design ensures that no critical stress exceeds the allowable value for any authorized combination of these parameters.

This proportionality is applicable during aerodynamic flight phases. It does not hold during non-aerodynamic phases such as takeoff roll, landing roll, or winch launch; these phases are therefore handled as non-instrumented operation and accounted at the reference EFI rate (see §4.2).

The Palmgren-Miner calculation is applied to stress amplitudes associated with load cycles, not to absolute static stress values alone. For a given critical location, if the stress amplitude is proportional to the load-factor amplitude within the authorized Utility-category flight envelope, the unknown proportionality coefficient is absorbed into the calibration of the EFI model and cancels when the reference and real spectra are compared using the same method. The absolute value of this coefficient is therefore not required for the proposed EFI-ratio approach. This does not mean that all structural locations have the same stress for a given load factor or share the same proportionality coefficient; it means that, within the authorized Utility envelope and for aerodynamic flight phases, load-factor variations provide a consistent proportional basis for the reference-versus-measured fatigue-severity comparison.

High Cycle Fatigue regime

Glider structures are designed so that operational stress levels remain well within the elastic domain of materials. The structure is therefore considered to operate in the High Cycle Fatigue (HCF) regime, which justifies the use of a Basquin-type *S/N* curve representation for the conservative EFI calculation used in this methodology.

4. Methodology

4.1 Reference Load Spectrum

The reference load spectrum for this methodology is the Kossira & Reinke Utility spectrum [2][8] (ABB.164). This is a deliberate initial-scope limitation: SLM is proposed at this stage only for gliders operated within Utility-category limitations. Aerobatic-category operation and any operation outside the Utility envelope are not credited under this methodology. The Utility spectrum is recognized by EASA (CM-S-006) as a basis for sailplane fatigue analysis. It is intentionally conservative: it covers the envelope of possible Utility-category glider usages and has built-in conservatism relative to typical operations.

The original Kossira & Reinke spectrum is published normalized to 6,000 flight hours. Because the certification EFI limit is anchored at the 12,000-hour life (§3), the published occurrences are scaled to the required exposure time before use in the EFI calculation. The original Kossira & Reinke spectrum is used rather than the more recent KoSMOS variant: KoSMOS omits low load factor variations to shorten ground-test duration, but those omissions are not appropriate when the reference spectrum is being compared cycle-for-cycle against a real measured spectrum.

4.1.1 Use of the reference spectrum in the methodology

The Kossira & Reinke Utility reference spectrum is used whenever EFI must be assessed against the certified reference basis. The detailed equations are introduced at the point where they are needed in the calculation:

1. The reference spectrum is scaled to the certified 12,000-hour life to anchor the EFI model to the certification EFI limit (§4.4).
2. Periods or phases without valid SLM data, including non-flight phases, are treated like non-instrumented operation and accounted at the reference EFI rate (§4.2 and §4.4).
3. For life-extension projection, the EFI produced by the reference spectrum is compared with the EFI produced by real flights over the same real-flight duration. Because this duration may be much smaller

than the duration of the published reference spectrum, the lower integer part of the scaled reference occurrences is used to keep the comparison fair and conservative (§4.5).

4.2 Real Load Spectrum Acquisition

The vertical load factor N_z is recorded continuously during flight using a tri-axial accelerometer located as close as possible to the Center of Gravity, in a rigid area near the wing spar. Proximity to the CG minimizes contamination from centripetal accelerations during attitude changes, while a rigid mounting area avoids local structural flexibility corrupting the measurement. Atmospheric turbulence and trajectory induce load variations at relatively low frequency, typically below 10 Hz (CS 22.341).

Accelerometer requirements:

- Minimum sampling rate: 20 Hz
- 6-face calibration at each annual inspection to ensure recorder performance
- Installation calibration with the glider in level-flight attitude and wings level, so that the recorder-corrected vertical axis is aligned with the gravity vector in the reference flight attitude

Data is stored on onboard electronic memory and uploaded to a central repository.

Non-flight phases. During takeoff roll, landing roll, winch launch, and any other phase dominated by non-aerodynamic forces, the proportionality between vertical load factor and structural stress no longer holds. The accelerometer signal from these sequences shall therefore not be used to build the real aerodynamic load spectrum. These sequences are treated like non-instrumented operation and are accounted at the reference Kossira & Reinke EFI rate corresponding to the Utility reference spectrum. This keeps the EFI accounting conservative and avoids using accelerometer data outside the domain in which N_z is a valid stress proxy.

4.3 Load Spectrum Processing — Kossira & Reinke Counting Method

Both the reference and real spectra are processed using the same Kossira & Reinke counting method [2] [8], ensuring a consistent and legitimate comparison. The method is described in Appendix A and summarized as follows:

1. Acquisition at 2^{10} resolution, corresponding to 1024 classes over a load factor range from -4g to +6g.
2. Hysteresis filtering.
3. Conversion to 2^5 resolution, corresponding to 32 load factor classes.
4. Identification of successive peaks and valleys.
5. Recording of transitions between consecutive peaks and valleys in a 32×32 Markov transition matrix.
6. Transformation of the matrix to count exceedances above and below the mean load factor, producing the load spectrum: occurrences as a function of load factor level.

The use of the same counting method for both spectra is essential to the validity of the comparison. The choice of counting method affects the resulting occurrences; consistency eliminates this as a source of discrepancy.

4.4 Equivalent Fatigue Index — Palmgren-Miner Rule

The Palmgren-Miner rule provides a linear cumulative index for fatigue. In classical notation this value is often denoted D , for damage. In the present SLM method, the same mathematical quantity is referred to as the **Equivalent Fatigue Index**, noted EFI , to avoid implying that the calculation represents measured physical damage in the composite structure:

$$EFI = \sum_i \frac{n_i}{N_i}$$

where n_i is the number of cycles at load factor level i , and N_i is the number of cycles to failure at that level.

For SLM, the Palmgren-Miner rule is used primarily as a consistent comparison framework between two spectra: the reference spectrum and the measured real-flight spectrum. The absolute value of EFI should not be interpreted as a direct measurement of physical composite damage at a specific structural location.

N_i values are derived from the reference spectrum and conservative fatigue parameters using the Basquin model. To simplify the notation, the positive EFI exponent is defined as:

$$k = -\frac{1}{b}$$

For the generic SLM methodology, the selected composite fatigue parameter is:

$$b = -0.16$$

and therefore:

$$k = 6.25$$

The selected value is a conservative engineering parameter for ageing composite glider primary structures made predominantly from conventional GFRP/CFRP epoxy laminates. It is not intended to be a universal composite-material constant or a type-specific material property.

The value is selected from the supporting review of published fatigue-fit parameters for epoxy-based GFRP/CFRP composite structures. Wind-turbine blade composite data are used as the closest available published analogue because they involve large, lightly loaded GFRP/CFRP structures operating in high-cycle fatigue under variable-amplitude loading. In the SNL/MSU/DOE fatigue database review, the steepest identified epoxy GFRP stress-fit exponent among the retained primary comparison data is $B = b = -0.1556$ for a QQ1 glass/epoxy multidirectional laminate at $R = 0.1$ [9]. Additional SNL/MSU/DOE epoxy GFRP data and AIAA/SNL-MSU-DOE database trends give less severe values, while carbon-dominated epoxy laminates show much shallower stress-life slopes [9][10]. General composite fatigue references also emphasize that fatigue parameters depend on material system, layup, stress ratio, manufacturing process, environment, defects, repairs, and failure mode [11][12][13].

The generic value $b = -0.16$ is therefore chosen because it is slightly steeper than the steepest retained GFRP epoxy value and remains strongly conservative for CFRP-dominated epoxy structures. The selected value shall be treated as a conservative EFI severity-weighting parameter for the generic SLM methodology. For a type-specific certification program, it should be confirmed by the certification authority and, where required, by representative material, detail, repair, environmental, or structural evidence.

The N_i expression, calibrated against the certification EFI limit, is:

$$N_i = C' |LF_i - LF_{average,ref}|^{-k}$$

where the calibrated constant C' is:

$$C' = 3 \sum_j n_{j,ref,12000} |LF_j - LF_{average,ref}|^k$$

where the symbols are defined as follows:

- i — index of the load factor level at which N_i is being evaluated.
- j — summation index running over all load factor levels of the reference spectrum. It is used inside the constant C' to distinguish the summation over the whole spectrum from the single level i at which N_i is evaluated. The two indices range over the same set of load factor classes; different letters are used only to avoid confusion between the bound summation variable and the free index i .
- $n_{j,ref,12000}$ — number of cycles (occurrences) at load factor level j in the reference Kossira & Reinke Utility spectrum, normalized to the certified life of 12,000 flight hours and processed with the counting method of §4.3.
- LF_j — load factor value of class j ; LF_i — load factor value of class i .
- $LF_{average,ref}$ — average load factor of the reference spectrum used to calibrate C' .
- $LF_{average,real}$ — average load factor of the valid real measured spectrum used to compute real EFI.
- b — Basquin S/N slope exponent; $b = -0.16$ for the selected generic composite case.
- $k = -1/b$ — positive EFI exponent used in the EFI summations; $k = 6.25$ for the selected generic composite case.
- C' — proportionality constant linking load factor amplitude to the number of cycles to failure. It is not a free parameter: it is fixed by calibrating the reference-spectrum EFI against the certification EFI limit $EFI_{ref,12000} = 1/3$.

Because the same calibrated constant C' , EFI exponent k , load-factor class definition, and Kossira & Reinke counting method are applied to both the reference and the real spectra, the absolute value of the load-factor-to-stress proportionality coefficient cancels in the EFI ratio and need not be known. The amplitude term is evaluated relative to the average load factor of the spectrum being assessed: $LF_{average,ref}$ for the reference spectrum and $LF_{average,real}$ for the real measured spectrum.

Real-flight EFI accumulation is then:

$$EFI_{real} = \frac{1}{C'} \sum_i n_{i,real} |LF_i - LF_{average,real}|^k$$

For gliders instrumented partway through their life, total EFI is:

$$EFI_{total} = EFI_{not\ instrumented} + EFI_{real}$$

where $EFI_{not\ instrumented}$ is computed using the reference spectrum normalized to the duration or operational exposure for which valid SLM data is not available, including non-flight phases treated as non-instrumented operation. EFI_{real} requires no normalization as it already corresponds to the valid instrumented aerodynamic flight period. For a glider instrumented late in life, $EFI_{not\ instrumented}$ can dominate EFI_{total} , which is why early instrumentation is advantageous (§4.5).

4.5 Life Extension Assessment

The EFI ratio R quantifies the life extension potential:

$$R = \frac{EFI_{ref}}{EFI_{real}}$$

where both values are computed over the same observed flight duration t_{real} . Here EFI_{ref} is the reference-spectrum EFI normalized to t_{real} — not the full-life value $EFI_{ref,12000} = 1/3$ of §3 and §4.4 — and EFI_{real} is the measured EFI of §4.4 over the same t_{real} .

To evaluate R in practice, the reference and real spectra must cover the same exposure time. The published reference spectrum (normalized to 6,000 flight hours) is scaled down to t_{real} , and the floor function $\lfloor \cdot \rfloor$ is applied to the scaled reference occurrences to add conservatism (the real observation period t_{real} being much shorter than 6,000 hours). The operational expression, using the EFI exponent k , is:

$$R = \frac{\sum_i \left\lfloor \frac{n_{i,ref} t_{real}}{6000} \right\rfloor |LF_i - LF_{average,ref}|^k}{\sum_i n_{i,real} |LF_i - LF_{average,real}|^k}$$

where $n_{i,ref}$ and $n_{i,real}$ are the occurrences at load factor level LF_i in the reference and real spectra respectively. This is the same computation as §4.4: substituting the EFI_{ref} and EFI_{real} expressions into $R = EFI_{ref}/EFI_{real}$, the constant C' is identical for both spectra and cancels, leaving only the spectra and the EFI exponent k .

The estimated life extension beyond the certified limit is:

$$t_{extension} = (R - 1) \times t_{remaining\ ref}$$

where $t_{remaining\ ref}$ is the flight time remaining until the certified 12,000-hour limit is reached under reference spectrum assumptions.

As an illustration, preliminary analysis of approximately 35 hours of flight data on an instrumented Janus B yielded an EFI ratio of about 26 for the selected composite parameter $k = 6.25$, corresponding to a potential life extension of several thousand hours for typical training and cross-country operations. In the supporting white paper [7], this illustrative case corresponds to approximately 7,500 hours of extension for an aircraft with about 300 hours remaining to the certified 12,000-hour limit. The same data evaluated with a metallic-part parameter $k = 5$ yields an EFI ratio of about 12, which may be useful when assessing the adequacy of continuing inspection intervals for metallic fittings and attachments. These figures are illustrative of the method rather than certified results.

Life extension is conditional on the assumption that future operations follow the same load pattern as those already observed. R and $t_{extension}$ are therefore recalculated at each Airworthiness Review Certificate (ARC) renewal based on accumulated data. R is highly sensitive to occasional high-load maneuvers: as an order of magnitude, a single 4g maneuver every two flight hours can reduce R by more than an order of magnitude relative to gentle cross-country flying.

5. Applicability and Limitations

- Identical to all other methods for fatigue life identification, this method is valid only when the glider is operated within its certified Utility-category flight envelope (speed, configuration, mass distribution, load factor, etc.).
- Aerobatic-category operation and any operation outside Utility-category limitations are excluded from the initial scope of SLM implementation.
- SLM is not certification by analysis, does not determine the absolute physical damage state of the composite structure and does not identify the critical structural location or critical failure mode.
- SLM does not reopen/change the Type Certificate structural substantiation, it uses the approved certification life exposure as the conservative reference limit.
- SLM compares measured operational severity against this accepted reference severity.
- The Palmgren-Miner calculation is used as a common comparison framework for both spectra.
- The Basquin parameter $b = -0.16$ is used as a conservative severity-weighting parameter, applied consistently to both the reference and measured spectra. It should not be considered a universal GFRP/CFRP epoxy composite-material property.
- Missing or invalid flight data as well as Non-flight phases (e.g. takeoff roll, landing roll, winch launch, and any phase dominated by non-aerodynamic forces), are treated like non-instrumented operation, accounted at the reference EFI rate (§4.2).
- The method remains dependent on Instructions for Continued Airworthiness (ICA): sensor calibrations, data processing, inspections, especially for defects, repairs, bonds, metallic fittings, and unforeseen local deterioration.

6. Concept of Operation associated with methodology

This section provides the methodology-level Concept of Operation (CONOPS) summary. The detailed CONOPS, release-to-service conditions, abnormal-case handling, and practical procedures are controlled by the STC certification package, in particular the Certification Programme, the CRI or equivalent means-of-compliance agreement, the Instructions for Continued Airworthiness (ICA), and the Installation, Operating and Maintenance Manual (IOMM) for the SLM recorder and associated data process.

The operational implementation shall be defined in those approved documents. They cover practical installation, calibration, data acquisition, data validation, processing, maintenance, inspection, reporting, and airworthiness decision rules. In particular, after operation beyond the certified safe-life limit has started, SLM availability, valid recorder calibration, validated EFI status reports, and the required inspections become conditions for continued airworthiness, as defined in the ICA and associated STC documents.

Phase 1 — Instrumentation and Data Collection (*below the certified life limit*)

The SLM recorder is installed in accordance with CS-STAN 104b / Form 123. Installation calibration and annual 6-face calibration are recorded in the aircraft logbook. Flight data are uploaded to the controlled SLM repository and processed using the approved toolset. Any flight or phase occurring without valid SLM data, including sensor fault, data corruption, missed calibration, or non-flight phases excluded from accelerometer-based processing, receives no measured-usage credit and is substituted at the reference EFI rate for the

corresponding duration or operational exposure, as defined in the approved ICA and data-processing procedure.

Phase 2 — EFI Evaluation at the Certified Life Limit

When the aircraft reaches the certified life limit, a dedicated structural inspection is performed in accordance with the approved ICA. The inspection covers the composite structure and the overall aircraft condition, including incidents, repairs, bonds, metallic fittings, and other items defined in the ICA. Continued operation may be authorized only if the inspection does not identify unacceptable findings and the approved EFI assessment shows:

$$EFI_{total} < \frac{1}{3}$$

Phase 3 — Operation Beyond the Certified Life Limit

Operation beyond the certified life limit is a continuation of Phase 1 monitoring with additional continued-airworthiness controls. A Go/No-Go decision is made at each ARC renewal or other approved review point using the validated SLM EFI status report. The Go criterion shall include both the current cumulative EFI and the projected EFI rate: EFI_{total} must remain below $1/3$, and the projected EFI rate must show that the $EFI = 1/3$ limit will not be reached before the next scheduled inspection or review. Recurring structural inspections continue at a maximum interval of 1,000 flight hours or 5 years, whichever occurs first. The process continues until the aircraft exhausts its remaining EFI potential or another approved continued-airworthiness limit is reached.

7. Approval and Deployment Pathway

This section gives only the methodology-level approval context. The detailed certification artifacts, document list, compliance responsibilities, EASA involvement, and approval schedule are controlled by the STC Certification Programme and associated Master Document List.

SLM deployment should be supported by an approved certification pathway, such as an EASA Supplemental Type Certificate (STC) or another approval route accepted by the competent authority. The Certification Programme defines the compliance approach for the STC, including the applicable aircraft, the approved operating domain, the proposed means of compliance, the required certification artifacts, and the EASA involvement for review or approval.

The certification package should include, as applicable:

- Certification Programme;
- Certification Review Item (CRI) or equivalent agreement on the proposed means of compliance;
- SLM Methodology Report (this document);
- Instructions for Continued Airworthiness (ICA);
- Installation, Operating and Maintenance Manual (IOMM) for the SLM recorder and associated data process;
- validation reports, Master Document List, Compliance Check List, and Master Compliance Report, as defined in the Certification Programme.

The Certification Programme shall remain the controlling document for the detailed approval artifacts. This methodology report defines the EFI calculation principles, assumptions, and limitations; the ICA and IOMM define the operational procedures, inspections, calibration, data handling, abnormal-case treatment, and release-to-service conditions.

8. Appendix A — Kossira & Reinke Processing Rules

This appendix summarizes the minimum processing rules required to implement the Kossira & Reinke counting method consistently for both reference and real spectra [2][8].

1. **Acquisition and fine resolution.** The acquired load factor signal is represented at 2^{10} resolution, corresponding to 1024 classes over a typical load factor range from -4g to +6g. This gives a fine resolution of approximately 0.01g.
2. **Hysteresis filtering.** A new sample is retained only if its difference from the last retained sample exceeds the Kossira & Reinke threshold DX , typically $DX = 10$ in the 1024-class representation, i.e. approximately 0.1g [8]. This prevents small oscillations near a class boundary from creating artificial fatigue cycles.
3. **Reduction to 32 classes.** Retained samples are converted to 2^5 resolution, corresponding to 32 load factor classes over the same range. The corresponding class width is approximately 0.32g.
4. **Peak and valley detection.** Successive local maxima and minima are identified in the filtered 32-class signal. Only alternating peaks and valleys are used for Markov transition counting.
5. **Markov transition matrix.** Each transition from one retained peak/valley value to the next increments the corresponding cell of a 32×32 Markov matrix.
6. **Exceedance spectrum.** The Markov matrix is transformed into occurrences above and below the average load factor. The resulting spectrum gives the number of occurrences associated with each load factor level.
7. **Consistency rule.** The same class definition, filtering rule, peak/valley rule, Markov convention, and exceedance transformation shall be used for the reference spectrum and for real measured data. This consistency is essential because different counting methods can produce different occurrence distributions.

Implementation of the method requires the applicable Kossira & Reinke Utility reference occurrence table. That table should be included in the approved calculation package in machine-readable form.

9. References

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